

**Dielectric constant (Dk).** The dielectric constant (Dk) of a PCB material is generally between 3.5 and 5.5. A material's Dk level will usually drop as the frequency rises. For a material to be safe for high-frequency applications, it must maintain a stable dielectric constant over the entire range of frequencies of operations.

The higher the relative dielectric constant, the slower a signal travels on a wire, the lower the impedance of a given trace geometry, and the larger the stray capacitance along a transmission line. Given a choice, the lower dielectric constant is preferable.

**Loss tangent.** A material's loss tangent gives a measure of the power lost due to the material. The lower a material's loss tangent, the better is the material.

The  $\tan \delta$  of most PCB materials ranges from 0.02 for the most used materials to 0.001 for very-low-loss, high-end materials. It also varies with frequency, increasing as frequency increases. Loss tangent is usually a critical consideration at frequencies greater than 1GHz.

**Glass transition temperature.** Glass transition temperature ( $T_g$ ) is the temperature range in which a PCB substrate transitions from a glassy, rigid state to a softened, deformable state as polymer chains become more mobile. When the material cools back down, its properties return to their original states.

**Decomposition temperature.** Decomposition temperature ( $T_d$ ) is the temperature at which a PCB material chemically decomposes (the material loses at least 5% of its mass). A material's  $T_d$  is an important ceiling when assembling PCBs, because when a material reaches or surpasses its  $T_d$ , changes to its properties are not reversible.

Choose a material where you can work in a temperature range that's higher than  $T_g$  but well below  $T_d$ . Most solder temperatures during PCB

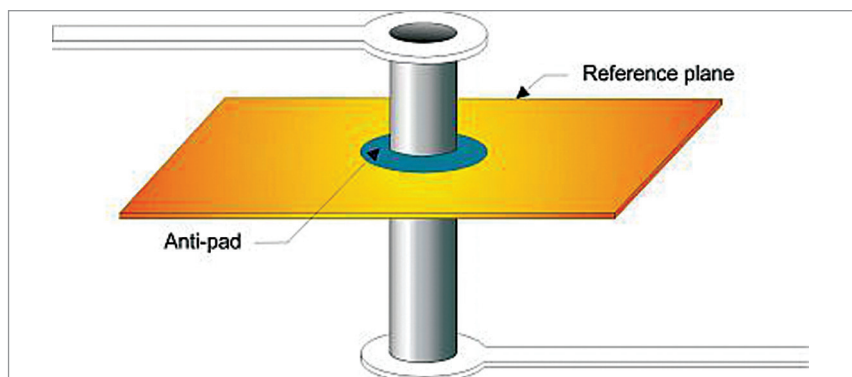


Fig. 5: Via anti-pad (Credit: <https://resources.pcb.cadence.com>)

assembly are in the 200°C to 250°C range and, luckily, most materials have a  $T_d$  greater than 320°C.

**Coefficient of thermal expansion.** The coefficient of thermal expansion (CTE) is the rate of expansion of a PCB material as it heats up. As a material's temperature rises past  $T_g$ , the CTE will rise rapidly.

The CTE of a substrate is usually much higher than copper, which can cause interconnection issues as a PCB is heated. CTE along the X and Y axes are generally low—around 10 to 20 ppm/°C. The CTE along the Z-axis should be as low as possible, and this will increase rapidly as a material surpasses  $T_g$ .

To reduce the higher expansion above  $T_g$ , choose a material with a higher  $T_g$ , at least 170°C and a low Z-axis CTE at a temperature greater than  $T_g$ .

### Some common mistakes to avoid

- Use proper trace width as calculated from the layer stack-up for the required characteristics impedance.
  - For power and ground traces and planes, the width must be sufficient to carry the current.
  - Place decoupling capacitors near/below the power pins.
  - Make separate grounds for analogue and digital signals.
  - Make sure the reference plane for critical signals is continuous end to end.

- Follow any special layout recommendations given by the IC manufacturer.

- Never route sensitive digital signals over high-voltage power planes.

- For differential signal length tuning, always match the asymmetric side.

- For high-speed signals, minimise the via stub—the unused via portion. For example, in a signal travelling from layer 1 to layer 18 in a 20-layer PCB, the part of the via in layers 19 and 20 is the stub. Back drilling may be done to remove the stub. But there is a cost penalty.

- Make sure power and ground planes expanding in multiple layers are well-stitched by vias. Also, do not place too many vias in a narrow area, which may, in turn, increase the current density and resistance.

- Keep a watch on the anti-pad. The shape of the anti-pad often affects high-frequency signals. SI analysis can suggest a suitable anti-pad shape for the design under consideration.

- Check no traces or planes are routed in the keep-out areas.

- Make sure the minimum distances specified during layout (for example, the spacing between via to via, via to pad, via to trace, etc) are manufacturable. Contact your PCB fabricator for their capability and cost beforehand.

Happy designing! **EFY**